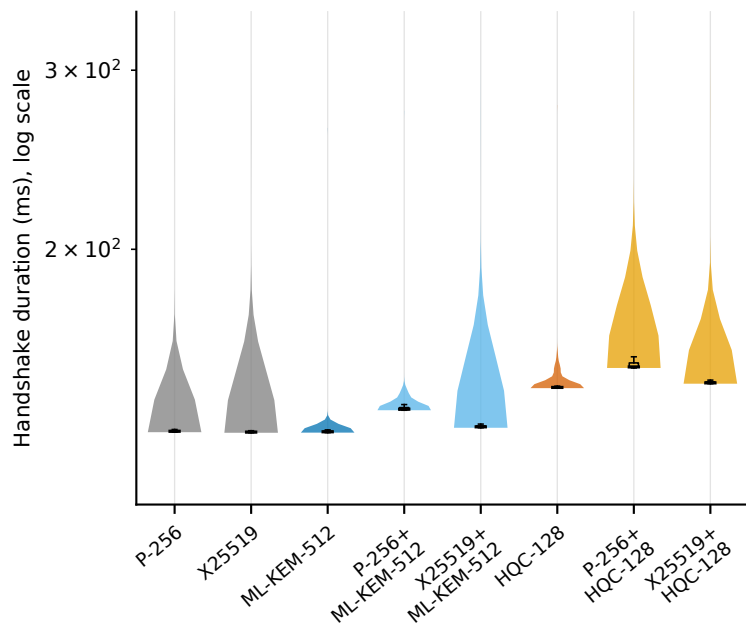
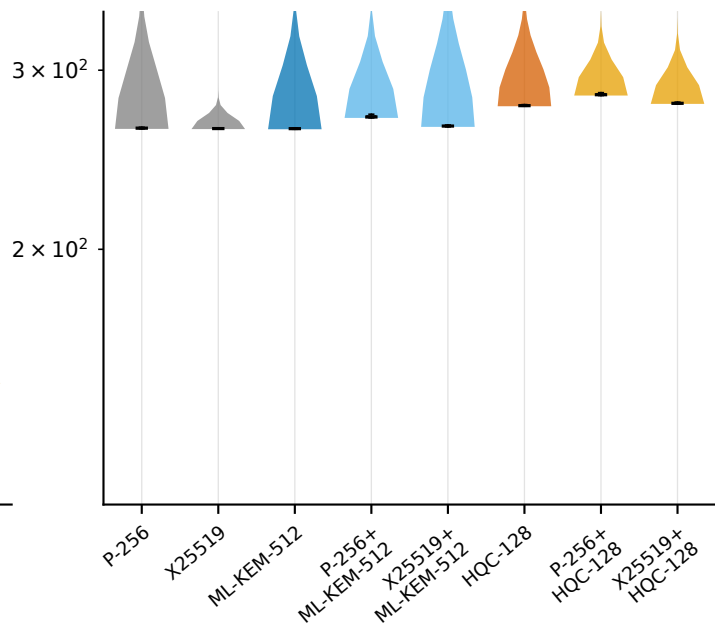


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**

Classical KEX ML-KEM (pure) ML-KEM (hybrid) HQC (pure) HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)
P-256	134.8	33.4	1.13k	271.9	44.4	2.14k
X25519	136.6	46.4	1.13k	264.1	8.5	747.2
ML-KEM-512	132.7	4.4	263.3	271.8	41.1	2.29k
P-256+ML-KEM-512	140.2	4.4	273.7	276.4	33.5	2.08k
X25519+ML-KEM-512	138.9	52.7	1.29k	275.6	50.8	2.07k
HQC-128	147.5	4.3	277.2	286.2	32.7	1.47k
P-256+HQC-128	160.4	45.4	1.30k	288.3	18.6	1.47k
X25519+HQC-128	151.6	34.3	1.30k	282.2	19.0	1.47k